Aneesh Prafull Kadam

+91 9004934077 | aneesh.kadam1@gmail.com | LinkedIn | GitHub | Thane, Maharashtra, India

Education

- **Sardar Vallabhbhai National Institute of Technology (SVNIT)** *Aug 2023 – Aug 2027*
  B.Tech – Electronics and Communication Engineering | CGPA: 6.23/10 Surat, India
- **Indian School Sohar** *2023*
  CBSE Class XII: 82% | Class X: 92% Oman

Relevant Coursework: Deep Learning, Digital Signal Processing, Linear Algebra, Probability & Statistics, Data Structures & Algorithms

Experience

- **OWL AI** *Dec 2025*
  AI/ML Intern Remote
  - Built and evaluated **4+ ML models** using Python, applying feature engineering and hyperparameter tuning to improve classification performance by **∼15%** over baseline.
  - Designed structured Jupyter Notebook pipelines for training, validation, and performance analysis across datasets with **50K+ samples**.
  - Maintained **5+ version-controlled repositories** on GitHub following collaborative branching and code review practices.

Projects

- **Speech Emotion Recognition – Multi-Scale CNN–Conformer Architecture** | Code *2026-27*
  Python, PyTorch, Librosa, Transformers | Dataset: RAVDESS
  - Built an end-to-end deep learning pipeline to classify **8 emotions** from raw speech: preprocessing → feature extraction → CNN → Conformer attention → classification.
  - Extracted **MFCC, Mel-Spectrogram, and Chroma** features to capture spectral, temporal, and tonal characteristics of speech signals.
  - Implemented **Conformer-based self-attention** to model long-range temporal dependencies, outperforming standard BiLSTM baselines.
  - Applied hierarchical capsule networks to improve emotion representation and reduce sensitivity to speaker variability.
  - Achieved **∼72% accuracy** on RAVDESS test split, surpassing CNN-only baseline by a significant margin.
- **Responsible AI Governance Framework – Loan Approval System** | Code *2026*
  Python, Scikit-learn, SHAP, Streamlit | Focus: Fairness, Explainability, Governance
  - Built an end-to-end Responsible AI system for loan approval on a **synthetic 2000-sample dataset**, integrating fairness, governance, and explainability layers beyond standard ML.
  - Implemented **SHAP-based explainability** to provide feature-level reasoning for every model prediction, enabling transparent and auditable decision-making.
  - Designed a **policy engine with RBAC** enforcing bias thresholds and confidence checks, routing decisions to ALLOW / REVIEW / REJECT levels.
  - Detected bias across gender subgroups using **demographic parity and equalized odds**; integrated Isolation Forest for real-time anomaly detection on suspicious predictions.

Skills

- **Programming:** Python, C, C++, SQL
- **Machine Learning:** Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Ensemble Methods, Anomaly Detection
- **Deep Learning / NLP:** Neural Networks, CNNs, Transformers, Conformers, Attention Mechanisms, NLP, Speech Processing, LLM Prompting
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, Hugging Face, SHAP, NumPy, Pandas, Librosa, Matplotlib, Seaborn
- **Tools:** Git, GitHub, Jupyter Notebook, VS Code, Streamlit, Supabase

Certifications

- **Machine Learning Specialization** *2025*
  DeepLearning.AI | Supervised ML · Advanced Algorithms · Unsupervised Learning, Recommenders & RL Coursera